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Dear Editor,

I kindly ask that you consider our manuscript, "*PoolParty: streamlined design of DNA sequence libraries in Python*," for publication as a Software article in *Genome Biology*.

Computationally designed DNA sequence libraries (oligonucleotide pools) are essential components of massively parallel reporter assays (MPRAs), protein deep mutational scanning (DMS) experiments, and other multiplex assays of variant effect (MAVEs). These libraries are also increasingly used in silico to probe the behavior of genomic AI models. Despite their importance, the process of designing these libraries remains tedious and error-prone. General-purpose toolkits such as Biopython can manipulate individual sequences but have no concept of a variant library as a structured object, while assay-specific tools such as VaLiAnT and MPRAnator are each tied to a particular experimental format. No existing tool provides a unified, composable framework for designing complex sequence libraries.

Our manuscript introduces PoolParty, an open-source Python package that addresses this need. In PoolParty, users build libraries by chaining together operations into a directed acyclic graph (DAG) using a concise, declarative syntax. This makes it possible to specify complex library designs—mixing systematic and random variation across multiple sequence regions—in just a few lines of code. Over 20 built-in operations cover nucleotide- and codon-level mutagenesis, motif insertion, deletion and insertion scans, barcode generation, and more. Importantly, PoolParty automatically generates "design cards" that record the specific operations used to construct each sequence, enabling design parameters to serve directly as covariates in downstream analyses. We demonstrate PoolParty's broad utility through three applications: a DMS library for protein GB1 comprising over 557,000 variants, an MPRA library probing transcriptional regulatory grammar, and an in silico surrogate modeling study of SpliceAI in which PoolParty's design cards provide the covariates for interpretable analysis of the deep learning model's behavior.

My lab has a track record of building and maintaining robust and highly cited open-source software, including Logomaker (Tareen and Kinney, *Bioinformatics*, 2020; **550 citations**) and MAVE-NN (Tareen et al., *Genome Biology*, 2022; **83 citations**). We have similarly taken care to make PoolParty robust, user friendly, open source, easily installable via pip, and well documented, with tutorials and comprehensive documentation hosted on ReadTheDocs.

We suggest the following experts as potential reviewers:

- Barak Cohen, Washington University in St. Louis, cohen@wustl.edu
- Ben Lehner, Wellcome Sanger Institute, ben.lehner@sanger.ac.uk
- Max Staller, UC Berkeley, mstaller@berkeley.edu
- Georg Seelig, University of Washington, gseelig@uw.edu
- Anshul Kundaje, Stanford University, akundaje@stanford.edu

My coauthors and I look forward to hearing back from you.

Sincerely,

Justin Block Kinney

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